

EXPERIMENT 11

Network Port-Scanning Detection

AIM

Analyze connection records and identify sources contacting unusually many destination ports or target systems.

ALGORITHM

1. Start.
2. Read/accept the required evidence data.
3. Process and analyze the data according to the experiment.
4. Apply the required threshold/rules/comparison.
5. Display the findings.
6. Stop.

CODE

```
from collections import defaultdict
r=[("10.0.0.5","192.168.1.10",22),("10.0.0.5","192.168.1.10",23),("10.0.0.5","192.168.1.10",80),("10.0.0.5","192.168.1.11",22),("10.0.0.5","192.168.1.12",443)]
p=defaultdict(set); t=defaultdict(set)
for s,d,x in r: p[s].add(x); t[s].add(d)
for s in p:
    print(s,"ports",len(p[s]),"targets",len(t[s]))
    if len(p[s])>=3 or len(t[s])>=3: print("Possible scanning behavior")
```

OUTPUT



```
PS: C:\Users\saich\AppData\Local\Programs\Microsoft VS Code\...
10.0.0.5 ports 4 targets 3
Possible scanning behavior
PS: C:\Users\saich\AppData\Local\Programs\Microsoft VS Code\...
```

RESULT

The program detected a source contacting multiple ports and systems, indicating possible scanning behavior.